

NAMAN YESHWANTH KUMAR

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SUMMARY

Full-stack engineer (MS Computer Engineering, ASU — GPA 3.81, May 2026) who solo-ships AI-powered products end-to-end. Built Trezzit (app.trezzit.com), a production React 19 + Django + Python SaaS serving **600+ users** for 12+ months — **341 components, 60+ REST endpoints, 995 tests**, multi-provider AI pipeline (Gemini / OpenAI / Claude / Grok) iterated from **60% → 95%+** accuracy through semantic validation and model-upgrade fallback. Merged **PR #655** into Intel's official self-checkout reference architecture.

TECHNICAL SKILLS

Full-Stack & TypeScript: Python | TypeScript | React 19 | Next.js | Tailwind CSS | Material-UI | Node.js | Django REST Framework | FastAPI | REST API Design | WebSocket Streaming | PWA

AI & LLM: LangGraph | LangChain | MCP Protocol | Tool Use / Function Calling | Multi-provider LLM Routing (Gemini / OpenAI / Claude / Grok) | RAG Pipelines | Streaming UIs | Pydantic Validation | Cursor | Claude Code

Backend Infra, Security & Testing: PostgreSQL | Redis (caching + rate limiting) | Celery (background jobs) | Cookie JWT + Google OAuth + OTP | Optimistic Locking | Encrypted Backups | Docker | AWS | Vercel | Nginx | GitHub Actions CI/CD | 995 Automated Tests | Linux

PROJECT EXPERIENCE

Trezzit — Production React + Django SaaS, Solo End-to-End (app.trezzit.com)

Feb 2025 – Present

- Solo-shipped Trezzit (app.trezzit.com) front-to-back — **341 React 19 components, 14 contexts (no Redux), 8 custom hooks**, Material-UI + Tailwind PWA with VAPID push and Share Target API; Django 5.1 + DRF backend with **60+ REST endpoints, 40 DB models, 995 tests**; PostgreSQL (Supabase), Redis, Celery + Celery Beat + Flower; DigitalOcean VPS (prod) + AWS Lightsail (staging) + Vercel; GitHub Actions CI/CD with prod→staging DB sync and encrypted backups.
- Built a two-stage AI pipeline routing across Gemini, OpenAI, Claude, and Grok — pass 1 + semantic validation gate + automatic model-upgrade fallback on failure — iterated parsing accuracy from **~60% → 95%+**; pre-computed balance system with O(1) reads, zero-sum invariant enforcement, and optimistic locking on bills; production telemetry tracks per-endpoint p95 latency and per-provider LLM tokens / cost.
- Security: cookie-based JWT with auto-refresh + failed-request queuing, Google OAuth via django-allauth, email OTP, rate limiting (100/hr anonymous, 10k/hr authenticated); built an MCP server letting AI assistants take actions across the platform (create bills, query balances, upload receipts) over the same auth and data layer as the web app.

SiliconCrew — Next.js Streaming Agent UI with 21 Tools & MCP Server

Nov 2025 – Present

- Built a Next.js 14 + TypeScript front-end with a resizable 3-panel layout (Sidebar | Chat | Artifacts), Monaco code editor, and **WebSocket-streamed tool-call cards** rendering in real time as the agent reasons over 21 tools; FastAPI + LangGraph backend with multi-provider LLM routing (Gemini / OpenAI / Anthropic) and full MCP server (stdio/SSE/HTTP) compatible with Claude Desktop and any MCP client; public at github.com/naman-ranka/siliconcrew.

HEAL.AI — AI Billing Assistant with RAG & Document Workflows

Sep – Nov 2025

- Won Devlabs Hackathon (1st place)** with HEAL.AI; built a RAG pipeline (PDF chunking + 768-dim embeddings + cosine-similarity retrieval) enabling natural-language Q&A, overcharge detection, and automated dispute-email generation via Gemini 2.5; FastAPI + Next.js + TypeScript, Dockerized.

Intel Automated Self-Checkout — Open Source Contribution

Jan – Feb 2025

- Contributed sensor analytics microservice to **Intel's official self-checkout reference architecture** — configurable data pipeline (MQTT, Kafka, HTTP) for real-time LiDAR and weight sensor data with Grafana dashboard; **PR #655 merged upstream** after Intel engineering review.

WORK EXPERIENCE

Software Developer (Sole Engineer)

Oct 2022 – Apr 2024

NCR GOLD

Bengaluru, India

- Engineered a production Django platform (PostgreSQL + Celery + Nginx on DigitalOcean) processing **18,000+ orders over 3 years**; built 3 role-based portals (admin, artisan, customer) with multi-method auth — automated PDF invoice generation and WhatsApp Business API delivery on every state transition, eliminating **~20 min** of manual relay per order.
- Reverse-engineered third-party Jewely ERP (**zero public API**) through direct SQL schema analysis; built bidirectional bridge model syncing production weights and ERP status in real time; engineered inbound WhatsApp state-machine chatbot (JSON state per user, admin + customer flows) — full accountability loop across all three portals without disrupting legacy system.

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Expected May 2026

Arizona State University — Coursework: Artificial Intelligence · Foundations of Algorithms · Reinforcement Learning in Robotics · Perception in Robotics · Algorithm/Hardware Co-Design

Tempe, AZ

Bachelor of Engineering — Electrical and Electronics

August 2022

BMS College of Engineering

Bengaluru, India

AWARDS & HONORS

- Won **'Best Use of AI'** at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform generating personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for unique password generation.